

Research Gap Analysis

The reviewed literature demonstrates significant progress in applying Large Language Models (LLMs) to Infrastructure-as-Code (IaC) generation, security validation, automated repair, configuration validation, industrial control systems, and smart manufacturing. Existing studies have introduced effective approaches for Terraform generation, security-aware IaC development, technology-agnostic repair, Kubernetes misconfiguration detection, enterprise DevSecOps validation, Infrastructure-as-Code benchmarking, PLC code generation and verification, Industrial IoT security, and intelligent Industrial Control System (ICS) generation. Despite these advances, the research remains fragmented, with most solutions addressing only specific stages of the Infrastructure-as-Code lifecycle or individual industrial domains rather than providing a unified validation framework.

A major limitation observed across the literature is the lack of **end-to-end, multi-format Infrastructure-as-Code validation**. Several studies are restricted to a single IaC technology such as Terraform, Kubernetes, or AWS CDK, while PLC-focused research concentrates only on industrial controller programming. Although these approaches achieve high accuracy within their respective domains, they cannot jointly validate heterogeneous cloud infrastructures consisting of Terraform, CloudFormation, Kubernetes, AWS CDK, Serverless templates, and industrial automation components within a single intelligent workflow. This limitation reduces their applicability in modern smart-manufacturing environments where cloud infrastructure and operational technology coexist.

Another recurring limitation is the heavy dependence on **static analysis or offline evaluation**. Most frameworks perform syntax checking, security scanning, benchmark evaluation, or formal verification before deployment, while providing limited support for monitoring deployed infrastructure. Industrial IoT studies remain largely conceptual, and ICS-generation frameworks prioritize rapid deployment without validating the correctness or security of generated configurations. Consequently, existing approaches fail to correlate Infrastructure-as-Code definitions with real-time operational information such as deployment logs, runtime telemetry, cloud monitoring metrics, and configuration drift, reducing their effectiveness in dynamic cloud-enabled manufacturing systems.

The literature also reveals that current solutions generally address **isolated tasks** such as code generation, security assessment, automated repair, PLC verification, or compliance analysis independently. None of the reviewed studies integrates intelligent parsing, security assessment, deployment verification, configuration drift detection, runtime telemetry analysis, cost evaluation, and explainable recommendation generation through a coordinated multi-agent architecture. Furthermore, several frameworks require human intervention, rely on domain-specific knowledge bases, or depend on external verification tools, limiting automation and scalability in large-scale industrial cloud environments.

Therefore, there remains a clear need for a **Telemetry-Aware Multi-Agent Infrastructure-as-Code Assurance Framework** capable of supporting multiple Infrastructure-as-Code technologies while integrating static validation, deployment verification, runtime cloud telemetry, configuration drift detection, security assessment, and explainable LLM-based recommendations within a unified workflow. Such a framework would bridge the gap between cloud-native Infrastructure-as-Code management and smart-manufacturing environments, providing secure, scalable, and intelligent infrastructure assurance across both Information Technology (IT) and Operational Technology (OT) ecosystems.